

Question 1. (calculation)

a) Find the Fourier pair coefficients G_k of the following periodic function:

$$g(t) = \begin{cases} -1, & -\frac{T_0}{2} \leq t < 0, \\ 1, & 0 \leq t < \frac{T_0}{2} \end{cases} \quad (\text{and repeated}).$$

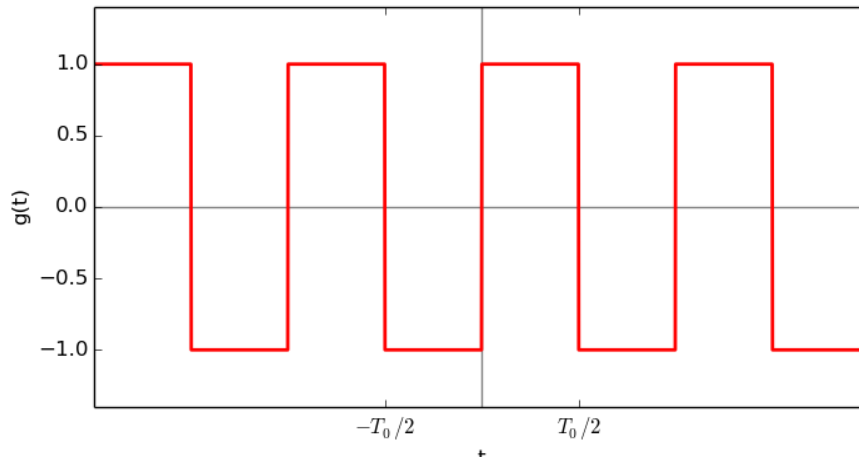


Figure 1: Function for Question 1.

b) Then, insert the spectral coefficients G_k that you have found *back* into the synthesis equation, so that $g(t)$ can be written as a sum of sinusoidal functions. Simplify as much as possible.

Question 2. (programming)

a) Use Python to plot the original piecewise function $g(t)$ from Question 1a (using the formula I've provided) for two periods (you will have to choose some value to be the period). Your plot should look like Figure 1, though maybe less wide. (You will have to pick some value for T_0 , but you should have a variable in your Python code to represent it still.) You will use the array of these values, below.

b) In Question 1b, you re-inserted your G_k values to find an expression for $g(t)$ as a sum of sinusoids. Let's investigate partial sum approximations to that expression (which probably has many terms).

Make an array of subplots, with 2 columns and 5 rows. Over the same t -interval used in question 2b, plot a copy of the following partial sums over k of the expression for $g(t)$ which you found (similar to what we did in class, using the following details):

Column 1:

- a) $k = 1$ (that is, just a single sinusoid)
- b) $1 \leq k \leq 5$
- c) $1 \leq k \leq 11$
- d) $1 \leq k \leq 25$
- e) $1 \leq k \leq 75$

Column 2:

f) $3 \leq k \leq 75$

g) $7 \leq k \leq 75$

h) $13 \leq k \leq 75$

i) $27 \leq k \leq 75$

j) $77 \leq k \leq 1000$

(If you wanted to, you could write a function that does the summation, with input arguments as lower and upper bounds of sums...)

Make sure you put a title on each plot so it is clearly defined.

What pattern do you notice as you go from ‘a’ to ‘e’? Which parts of the original function (the horizontal or the vertical ones) are relatively better represented by these, in general?

What pattern do you notice as you go from ‘e’ to ‘i’? Which parts of the original function (the horizontal or the vertical ones) are relatively better represented by these, in general?

Question 3. (calculation)

Find the Fourier transform $G(f)$ of the following (non-periodic) function:

$$g(t) = \begin{cases} 1, & -\frac{A}{2} < t < \frac{A}{2} \\ 0, & \text{elsewhere.} \end{cases}$$

(NB: A is **not** a period, it’s just a particular time value for the width of the support of this function.)

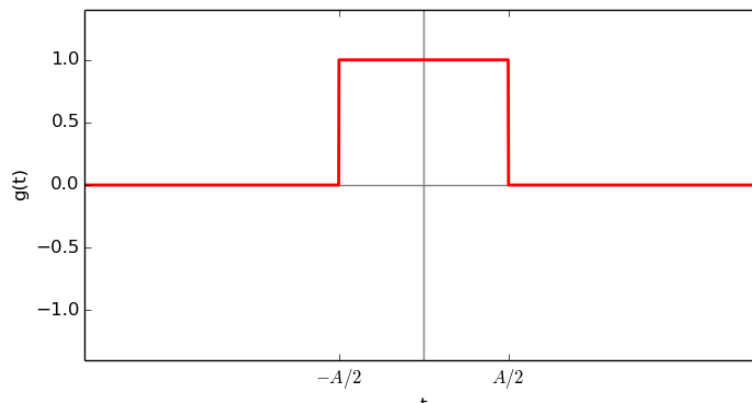


Figure 2: Function for Question 3.

Question 4. (programming)

This uses your results from Question 3. Make an array of subplots, with 2 columns and 1 row. The left plot should have “time” along the horizontal axis, so we can plot cases of $g(t)$ there. The right plot should have “frequency” along the horizontal axis, so we can plot $G(f)$ there. Plot the following 4 curves on each plot (and add labels to the plot with the “legend” functionality): where $A = 1, 2, 4, 8$; for each value of A use a different color, but $A = 1$ has the same value in each plot, etc. What pattern do you notice as A increases in each plot?

Note: use titles and axis labels in each subplot.

Bonus question. (calculation)

The mutual orthogonality of harmonic sinusoids within a period is an important property for us being able to have the Fourier relations. (That is, the sinusoids form an orthogonal basis.)

Show the values of each of the following integrals for the following three cases of integers p and q :

- $p = q = 0$
- $p = q \neq 0$
- $p \neq q$,

where $\omega_0 = 2\pi/T_0$ and T_0 is the period in question.

$$\int_{-T_0/2}^{T_0/2} \sin(p\omega_0 t) \sin(q\omega_0 t) dt \quad (1)$$

$$\int_{-T_0/2}^{T_0/2} \cos(p\omega_0 t) \cos(q\omega_0 t) dt \quad (2)$$

$$\int_{-T_0/2}^{T_0/2} \cos(p\omega_0 t) \sin(q\omega_0 t) dt \quad (3)$$

Hint: Using trigonometric identities might be helpful for simplifying integrals, like angle addition and subtraction formulas, and maybe even double angle formulas. But I've said too much...